Michael S. Bolton., MBA

michael.bolton.ph@dartmouth.edu
LinkedIn Profile URL: https://www.linkedin.com/in/bolton-2600/

- 15+ years across public-sector IT, healthcare, data, security, software, and regulated technology, including AI-enabled systems; 0-1 builder.
- Governed California county-health AI adoption across approximately 8 to 10 tools, using risk-register controls, NIST AI RMF, SIMM 5305-F GenAI and 5310-C privacy assessments, HIPAA/security analysis, and policy development to create traceable approval and oversight decisions.
- Deep AI/ML and LLM expertise spanning computer vision, predictive analytics, intelligent automation, generative AI, and agentic systems; independently evaluates architecture, data flows, APIs, model behavior, security boundaries, vendor claims, monitoring, and failure modes.
- Public-sector transformation leader from Deloitte AI & Data Engineering; federal and state engagements required mastery of sensitive financial workflows, legacy systems, business rules, dependencies, controls, and technical debt before future-state design, TDD, hands-on development, UAT, and CoE scale-up.
- Public-health and regulated-system leader who translates technical risk for executives, security, privacy, operations, and end users; applies health-equity, SDOH, assurance, and incident-review methods to make AI adoption safe, auditable, and operationally useful.

EDUCATION

Dartmouth College | Geisel School of Medicine, Hanover, NH **June 2027**
Master of Public Health (MPH) – Hybrid Program - Cohort Class of 2027
Applied Practice Experience **(APE) Leadership Toolkit Link:** https://www.pfactoolkit.org/

University of Wyoming, Laramie, WY **May 2024**
Master of Business Administration (MBA) General Management **GPA 3.7/4.0**
Honors/Awards: *Golden Key International Honor Society – Top 15%,*

Youngstown State University, Youngstown, OH **May 2008**
Bachelors Biological Sciences GPA 3.15/4.0
Honors/Awards: *Alpha Epsilon Delta Honor Society, Quest Scholar*.

WORK EXPERIENCE

Amygdal.ai Research, San Francisco, CA **February 2024-Current**
Founder & Build Engineer – Advanced AI Research & Agentic AI Systems | Data Engineering

- ♦ **Emporion Preview:** Live Site **GitHub:** mega-byte2600/hedge-desk

- Built Emporion from 0-1 as a Python/JavaScript multi-asset research platform, consolidating a seven-desk architecture and eight-view console with equities, options, arbitrage, dividend and event research, real CBOE option-chain analysis, read-only Schwab market-data APIs, scenario testing, Yellow Sheets, and deterministic real-estate IRR/XIRR/MIRR and sensitivity models into one repeatable research-to-decision workflow.
- Engineered production-grade assurance across the platform with 318 deterministic tests, 82% branch coverage, Python 3.9/3.11/3.13 CI, CodeQL, 72 war-game scenarios including 45 NO_TRADE controls, tamper-evident audit lineage, and automated paper evaluations every 15 minutes, replacing ad hoc model judgment; measurable release, risk, and review gates.

- ♦ **Hardball Dashboard:** Live Site **GitHub:** mega-byte2600/hardball

- Built Hardball from 0-1 as an MLB decision-intelligence platform spanning all 30 organizations and 16 seasons, integrating performance, payroll, 35,000+ injury/transaction events, free agency, MiLB, NPB, projections, health evidence, and market data into 25 traceable analyses for player evaluation, availability, roster construction, and budget decisions.
- Replaced heuristic roster construction with exact mixed-integer optimization, increasing projected roster value by +6.9 WAR at $171.2M, +7.1 WAR at $208.0M, and +2.5 WAR at $245.3M versus the benchmark build; stress-tested results across price, age, and availability assumptions.

County of San Luis Obispo (SLO Health), San Luis Obispo, CA **January 2026-October 2026**
Systems Administrator II (Information Security Analyst) – LIMITED-TERM

- Delivered in roughly 2 months the electronic incident-reporting capability as department was unable to implement over the prior 3 to 4 years, replacing paper forms and manual logs with centralized tracking, escalation, and follow-through.
- Evaluated and governed approximately 8 to 10 tools with AI components, documenting use, risk, controls,+ decisions in the organizational risk register; applying NIST AI RMF, HIPAA, privacy, & security criteria to create traceable oversight.
- For Eleos, completed SIMM 5305-F GenAI and SIMM 5310-C privacy assessments, independently examining application, data, model, API, security, and EHR behavior, challenging unsupported vendor claims, and authoring the initial AI use policy.
- Briefed CISO and IT leadership on LLM and agentic-AI threats, translating jailbreaks, prompt injection, poisoning, data exfiltration, and sandbox escape into audit, control, and executive risk language.

US Centers for Disease Control and Prevention (CDC), Atlanta, GA **September 2024-May 2025**

IT Specialist

- Modernized 250+ BSL-4 and CLIA laboratory endpoints to Windows 11, replacing fragmented deployment tracking with centralized status and exception dashboards and reducing the support backlog by 63%.
- Automated endpoint compliance and reporting with SCCM and Intune, cutting audit preparation from 4 hours to 30 minutes, an 88% reduction, while improving reporting consistency and operational visibility.
- Represented CDC technical interests across FDA, NIH, and ARPA-H programs, harmonizing public-health data pipelines and research-system integration to improve interoperability by 40%.

Abbott Laboratories, Santa Rosa, CA **March 2022-January 2024**

Operational Excellence Lead | Sr. CAPA Engineer R&D

- Reengineered an RT5 process across three CLIA toxicology laboratories using baseline metrics, root-cause analysis, workflow redesign, and sustained controls, reducing downtime and delivering approximately $250K in cost avoidance.
- Closed 47 sitewide CAPAs under CLIA standards through risk-based remediation, structured root-cause analysis, effectiveness checks, and quality metrics, strengthening safety and regulatory performance.
- Designed and facilitated Lean Six Sigma Yellow Belt training for cross-functional staff, building improvement capability and a shared operating discipline around metrics, root cause, and sustained control.

Deloitte, Lake Mary, FL **June 2018-August 2020**

Sr. Solutions Engineer

- Transformed federal and state agency workflows within Deloitte AI & Data Engineering by mastering sensitive financial processes, business rules, controls, dependencies, and legacy COBOL, Perl, and on-premises environments before translating current-state findings into future-state architecture and TDD components.
- Executed solutions from kickoff through UAT in 2 to 4 month Agile delivery cycles, coordinating roughly 6 to 10 offshore and 5 onshore developers and QA while remaining hands on across Python, AWS, APIs, UiPath, testing, and implementation.
- Scaled agency automation from individual workflows to governed platforms supporting avg. of 20 to 50 digital workers per engagement, then established Intelligent Automation CoEs with intake, prioritization, reusable standards, monitoring, exception handling, and change control.
- Modernized AWS data and integration foundations across ingestion, security, logging, APIs, and KPI reporting, reducing reliance on brittle legacy handoffs and creating supportable environments for analytics, automation, & model workloads.

Fresenius Medical Care, Concord, CA **September 2017-March 2018**

Sr. Verification Engineer Global R&D (*Contractor*)

- Validated the Versi PD Cycler end to end, coordinating software, hardware, risk, traceability, and regulatory evidence through launch with zero downtime.
- Automated regression and validation workflows with pytest, pandas, and SQL, reducing manual effort by 40% while improving repeatability, evidence quality, and audit readiness.

Zero Two Infinity Studios Inc., Berkeley, CA **April 2007-April 2018**

Co-Founder & President

- Co-founded and scaled a 0-1 cloud software company to 75+ clients, owning product strategy, technical delivery, customer adoption, and production operations; AWS architecture changes reduced streaming latency by 40%.

Philips Healthcare, Pittsburgh, PA **May 2016-May 2017**

Sr. V&V Engineer New Product Introduction (NPI) R&D (*Contractor*)

- Delivered Agile verification and risk execution for DreamStation Go CPAP during FDA 510(k) submission, coordinating cross-functional work and achieving a 98% first-pass rate.
- Built risk-based hardware/software traceability under ISO 14971, surfacing failure modes earlier and strengthening safety, validation, and defect-resolution decisions.

Auris Surgical Robotics, Redwood City, CA **August 2015-December 2015**

Sr. Test Engineer – Series B Funded Start-up (*Contractor*)

- Supported early Monarch robotic bronchoscopy prototypes using computer vision to map patient anatomy and enrich procedural context; generated verification and performance data that informed intended-use decisions and development of the emerging AI-enabled platform.
- Designed instrument-life, catheter, and mechanical performance testing with Python-supported analysis, using failure data to establish safe-use limits and objective development evidence for the computer-vision-enabled system.

LEADERSHIP AND SERVICE

United States Navy, Naval Air Station Moffett Field, Mountain View, CA **April 2002-April 2007**
USN Sailor – CTI3 – Cryptology Technical Interpretive

SKILLS & CERTIFICATIONS

Software Engineering: Python, Java, JavaScript, C#, .NET, SQL, REST APIs, GraphQL, Node.js, Angular, Kotlin, Android SDK, Xcode, Git/GitHub, Bash, Linux/RHEL, Trader APIs.

Cloud & Data Engineering: AWS, GCP, BigQuery, Databricks, Docker, GitHub Actions, Azure DevOps, ETL, feature engineering, Power BI, Tableau, WRDS, CMS VRDC.

AI/ML Engineering: NumPy, pandas, scikit-learn, XGBoost, TensorFlow/Keras, PyTorch, statistical modeling, regression, classification, clustering, time series, neural networks, model training/evaluation, RAG, Hugging Face, OpenAI API.

Multi-Model, Agentic & Adaptive AI: Hermes, Qwen3.5-9b, Qwen36-35B-A3B, Ornith-1.5, Ornith-1.5-9b, OpenAI Codex, Claude Code, GPT/Sol/Astra, Claude/Fable, DeepSeek V4 GLM 5.3 Flash, LongCat, LangChain, LangGraph, LangSmith, model routing, tool calling, context engineering, AI evaluation, model monitoring, deterministic validation, human approval controls. (BenchLM.ai,)

Security & Enterprise Systems: Splunk, Rapid7, Microsoft Entra ID, OWASP, IAM, least privilege, Linux hardening, SmartCare, Epic, HP ALM, DOORS, JIRA, Confluence, TrackWise, Notion.

Certifications:

- DeepLearning.AI | Stanford Online - Andrew Ng.
  - Machine Learning Specialization
- Andrej Karpathy Student
  - *Building makemore pt 1-4, Let's Reproduce GPT-2, Let's Build the GPT Tokenizer*
- AWS Certified Machine Learning Engineer
  - ** *In Progress** TBD exam date Spring 2027 MLA- CO2 -2027
- AWS Certified Solutions Architect Associate – SAA-C03
- AWS Certified Developer Associate – DVA-C02
- Dartmouth Committee for the Protection of Human Subjects. Collaborative IRB Training Initiative (CITI):
  - Group 1: Biomedical Research & Group 2: Social/Behavioral Research
- Harvard Business School: Quantitative Methods Certification
  - Abbott Rapid Diagnostics Operational Excellence: Lean Green Belt Certification
- FINRA: Securities Industry Essentials (SIE)
- RABQSA: Certified Internal Auditor ISO 9001 | ISO 13485

Professional Associations:

- AI Alliance - Member
- SANS Institute - Developer
- Google Developer GEAR - Developer
- Apple Developer Program - Developer
- Schwab Developer Portal: Individual Developer
- Dartmouth College Hugging Face Organization - Developer
- Society for American Baseball Research (SABR) - Member

Languages: Spanish, French, Portuguese-Brasileira, Arabic, Tagalog